

Forecasting Rideshare Demand at 1–6 Week Horizons

A few pieces of advice

Where this comes from. These notes are distilled from a detailed study of *week-ahead highway-traffic* forecasting. Rideshare demand shares its skeleton — a dominant weekly $\times$ daily seasonal cycle driven by human schedules, strong holiday/event sensitivity — so the structural lessons transfer. But rideshare differs from highway traffic in ways that *change where modeling pays*, and the 1–6 week horizon is its own regime. Here’s what I’d tell you.

1 Start here (and don’t skip it)

1. **Build the calendar-adjusted seasonal baseline first, and make it the bar everything must clear.** Per zone, the rule is:

$$\text{forecast} = (\text{recency-weighted average of the same day-of-week-and-hour over the last few weeks}) \times (\text{event/holiday factor on special days})$$

The seasonal average uses geometrically-decaying weekly weights (one parameter); the factor is the historical ratio (actual / baseline) on that holiday/event, pooled over past occurrences. On highway flow this reached $R^2 \approx 0.95$ and a weekly-aggregate error around 3.5%, and *nothing fancier beat it* — autoregression, gradient boosting, pooled panel models, cross-unit features all tied or lost. Expect a similarly brutal baseline for rideshare. Anything you deploy must beat *this*, in a leakage-free rolling backtest, before it earns its complexity.

2. **Know which “band” 1–6 weeks lives in.** Write demand as *level $\times$ seasonal shape + noise*. The forecast horizon is a low-pass filter: as it grows, the short-range, spatial, surge-propagation signal (what real-time dispatch models exploit) decays away. By 1–6 weeks you are forecasting the *seasonal backbone plus a slowly-moving level* — almost none of the minutes-ahead structure survives. Practical consequence: **do not reach for the spatio-temporal deep nets built for short-horizon dispatch**; they shine at minutes, not weeks. Different horizon, different toolbox.

2 Where a learned model genuinely helps at 1–6 weeks

The honest framing: at this horizon a model cannot extract hidden signal from demand’s *own past* — the seasonal baseline already captures it. Its value is in *learning a known response* and *reducing variance*. Rideshare, happily, has far more of both than highway traffic.

3. **Your #1 lever: learn the known-ahead event & calendar response.** Rideshare is much more event-driven than traffic — concerts, sports, conventions, festivals, holidays, New Year’s Eve, airport surges. The killer property is that these are **known weeks ahead**, so they need

no covariate forecast. Generalize the hand-crafted holiday factor into a *learned* event-response model (gradient boosting or hierarchical) with event metadata (venue, capacity, type) interacted with zone and time-of-day, plus per-zone heterogeneity. This is the clearest place a learned model beats the plain baseline at this horizon.

4. **Model the level with macro / known covariates — it is more dynamic than traffic’s.** Highway traffic is a near-utility with a near-frozen level; rideshare is a *discretionary* service whose level moves with employment, tourism (flight arrivals, hotel occupancy), fuel prices, competitor presence, marketing, and your own adoption curve. That level is partly forecastable from slowly-varying or known-ahead covariates. A structural / state-space model (level + trend + seasonal + regression), or a global model with these inputs, is the right tool — and it matters *increasingly toward 6 weeks*, where level staleness, not the seasonal shape, dominates the error.
5. **Exploit your panel — pool across zones and cities.** You almost certainly have far more cross-sectional units than a sensor network (zones, hexes, cities) plus coupled series (demand, supply, ETA). A single *global* model (DeepAR / temporal-fusion-transformer / global gradient boosting) with zone embeddings and shared covariates pools strength: it denoises sparse, low-demand zones and learns the event/calendar response far better than per-zone fits. This is where “more data” genuinely pays — concentrated on the *noisy-zone tail* and the *shared response function*, not on mining each zone’s own history.
6. **Forecast distributions, not just point means.** Supply positioning, driver incentives, and surge decisions care about the *spread and the tails*, not the average. Even where your point forecast only ties the baseline, calibrated quantiles — especially event-driven and regime-dependent uncertainty — are high-value and genuinely model-amenable. Train on quantile/pinball loss and report calibration, not just MAPE.

3 Traps that will quietly cost you

7. **Respect the known-ahead constraint — and the surge-leakage trap.** At 1–6 weeks you may only use covariates whose *future* value is known at inference. Events, holidays, calendar, scheduled promotions and planned price changes: *yes*. **Weather: no** — you cannot forecast it 1–6 weeks out; at best fold in climatology, and never train on weather *actuals* you won’t have in production. **Price / surge is endogenous:** if you train demand on realized surge you leak badly (you won’t know future surge, and surge itself responds to demand). Model demand at a reference price, or treat *your own planned* pricing as a known input. This leakage silently inflates backtests and breaks on deployment.
8. **Assume the seasonal shape drifts.** Traffic’s weekly shape is frozen year to year; rideshare’s is not (adoption, behavior change, work-from-home shifting commute ↔ leisure mix). Use *adaptive* seasonality (EWMA over weeks, or a time-varying seasonal), refit on a schedule, and monitor for drift and regime breaks (a pandemic, a city launch, a competitor’s exit). A static seasonal profile will slowly rot.
9. **Get the evaluation right — this is where most effort is wasted or misled.**
 - Proper *rolling-origin* backtests, with the known-ahead constraint enforced (parameters fit only on data prior to each origin).
 - Beware **MAPE on low-demand zones:** tiny denominators explode it (we saw >500% on quiet hours that were forecast fine in absolute terms). Aggregate, or use weighted / scaled

/ quantile metrics.

- **Bucket** the errors — event vs. normal days, by zone density, by regime. The headline number hides exactly where you win and lose.
- Always report **skill over the baseline** (item 1), not raw error.
- Estimate the **predictability ceiling** (e.g. an oracle in-sample fit) so you know how much headroom exists before you chase it.

4 The mindset

10. **Complexity should follow information, not precede it.** A transformer will not beat a calendar average on the deterministic backbone — that part is solved. Spend model complexity only where rideshare carries exploitable information *beyond* the regular pattern: events, the macro-driven level, panel pooling, and the decision-relevant tails. The honest sequence is always: *nail the baseline* → *measure where the variance and the headroom actually are* → *point the learned model exactly there*.

The good news. Highway traffic was the extreme case — almost all deterministic backbone, little else — which is why a one-parameter rule was unbeatable. Rideshare moves a lot of mass into the components where models earn their keep: events, macro coupling, a large panel, regime change, and a decision-coupled objective. So a well-aimed learned model has *real* room here. The whole trick is that it be *aimed*, not merely deployed.